

BIG DATA ANALYSIS

GROUP PROJECT

DUE 5TH JUNE, 2026 23:59H

As you will all be finishing your degree and entering the job market we will simulate a real-world data science project for your assessment. This is a group project, and you should form groups of **4-5 people**.

Please read the instructions and grading criteria carefully.

Big Data Analysis in Spark

You and your group are working on a data transformation project at BigDataCompany. BigDataCompany is a consulting firm that helps client organizations solve big data challenges, such as scaling ETL pipelines, improving data quality, enabling faster analytics, and operationalizing machine learning. For this project, you will act as the consulting team delivering an initial prototype and management-ready findings for an imaginary client company. The client has data that they want to process at scale, and they are considering PySpark as the core technology for their solution. It is up to your group to demonstrate:

- how PySpark can be used effectively for the client's data and use cases, and
- why Spark is valuable (in a big data context) for the tasks you choose to implement.

Your task is then as follows:

- 1) Find **up to** three datasets which you will explore with PySpark. Choose a dataset that is both challenging and interesting. Public repositories like UCI Machine Learning Repository, Kaggle, and governmental databases are excellent places to start.

Please note: The three datasets don't need to relate to each other. The idea is that you will use different functionality in Spark, and one dataset might be good for machine learning and the other better for graph analysis.

Please note: The selection of dataset is at your discretion. As such, selecting a dataset that requires minimal preprocessing does not in itself justify the omission of said pre-processing tasks from the submitted work.

With this dataset (or datasets), you will demonstrate your skills as PySpark data scientists and engineers working on a client delivery.

- 2) To do so you should generate interesting insights from the data and demonstrate your ability to build useful tools with the data (for forecasting, classifying, etc.) The more ability

you can demonstrate across the data scientist tasks, the better (ETL, data cleaning, exploration and summarising, visualisation, machine learning, data engineering, data streaming, graph analytics, etc.).

- 3) **Most importantly**, demonstrate your ability to understand big data and use PySpark correctly. This means demonstrating your ability to work with the functionalities seen in class and showing that you understand what Spark does well, and why it's valuable for large-scale processing.

You should (at the least) aim to show the following:

- **RDDs**
- **DataFrames & SparkSQL**
- **Pipelines & Data Engineering.**
- **Machine Learning (Spark Mllib), Deep Learning.**
- **GraphFrames.**
- **Code Quality and Documentation** – readable, well-commented code, clear structure, and meaningful variable names.
- **Streaming (Bonus) – Not required but including this will make your group eligible for bonus points.**

Important - It also means being very careful, you won't be working with big data, but you should be careful to ensure your code would work if your database were huge and spread across multiple partitions. Think about the ordering of data returned from queries and the libraries you use. **You should indicate anywhere explicitly that you have not used big data safe functionality, and why.**

You will need to deliver:

- A. You will prepare a **a PowerPoint presentation** aimed at management, meaning a non-technical audience. It should communicate the value of your work for the client: the insights you identified, the prototype tools you developed, and why Spark is an appropriate solution for the client's scale and needs. The presentation should not be technical. It should be framed as a consulting update, for example: "Here is how the client can make use of big data with Spark for these datasets and business questions," rather than: "Here are the model parameters," or "We used algorithm Y with hyperparameter settings Z." Keep the focus on Spark's role, the nature of the data challenges (think about the 4 V's), and what the client gains by using Spark.

- The presentation should be **no more than 10 slides total**, including the title, references, and other slides.
 - You must use the official **NOVA IMS PowerPoint template**.
 - The goal is to communicate insights and Spark's usefulness to management clearly, avoiding jargon, while still showing that you understand the technical work behind it.
- B. A **single ZIP file** submitted through Moodle. This ZIP must contain:
- All your **executed** PySpark notebooks (.ipynb) with visible outputs.
 - This is the only code we will consider. You can include .py files (or other files) that you use to save any classes or objects you create to use in your project.
 - Your **PowerPoint presentation exported as a PDF** (non-editable format).
 - **We will not grade any other formats.** Do not submit .html, .pptx, Google Drive links, or notebooks with no outputs. **Notebooks without outputs will not be graded.** Points will be deducted for late submissions.

We will hold online presentations and interview sessions. These will be held in the week following the submission deadline. On Moodle, a link to choose your groups will be set up. **Use this to assign your group members to a group number.** Once the internal dates are confirmed, we will use the Moodle group numbers to assign each group a slot. **If your group has not signed up on Moodle, they will not receive a slot and consequently will not be given a chance to present your work. This can result in a point deduction.** During your presentation, you will present your slides and answer questions about your notebooks and modelling choices. Difficulty explaining your own work will negatively affect your grade. Each group will be given a 15 minute slot, where they have **5 minutes** to present, and another 5 minutes to answer any questions we might have.

Remember that your grade depends primarily on your understanding and correct use of Spark! Demonstrate as many relevant functionalities as possible, keep your work clear and well-organised, and ensure your presentation shows that you understand both the technical and managerial perspectives of big data.

FAQ's

- **Can I have x number of people in my group?** Groups who do not have the correct number of people will receive a 1 mark penalty for each member under or over the limit.
- **Do I need to use a big database for this big data assessment?** No. Since we only have access to small resources, you are not expected to work with massive datasets.

However, your code and reasoning should still be applicable in a big data context.

However, all the analysis you apply should be applicable in the case that you are using a high volume of data. Hint: is the library you're using capable of dealing with big data?

- **Can we extend the deadline?** No.
- **What happens if I submit in the wrong format or with missing outputs?** We will not grade submissions that do not follow the required format or have missing outputs.

Please ask your practical and theoretical professors if you have any further questions.